Preliminary Design Masterplan

The room schedule, and how 15,000 m² is allocated

The masterplan is generated, not drawn. Every room is a box in the crescent's own polar frame, mapped to site metres and clipped to the boundary — so each area is the measured area of the polygon actually drawn, and the schedule closes on 15,000 m² without anyone reconciling a spreadsheet.

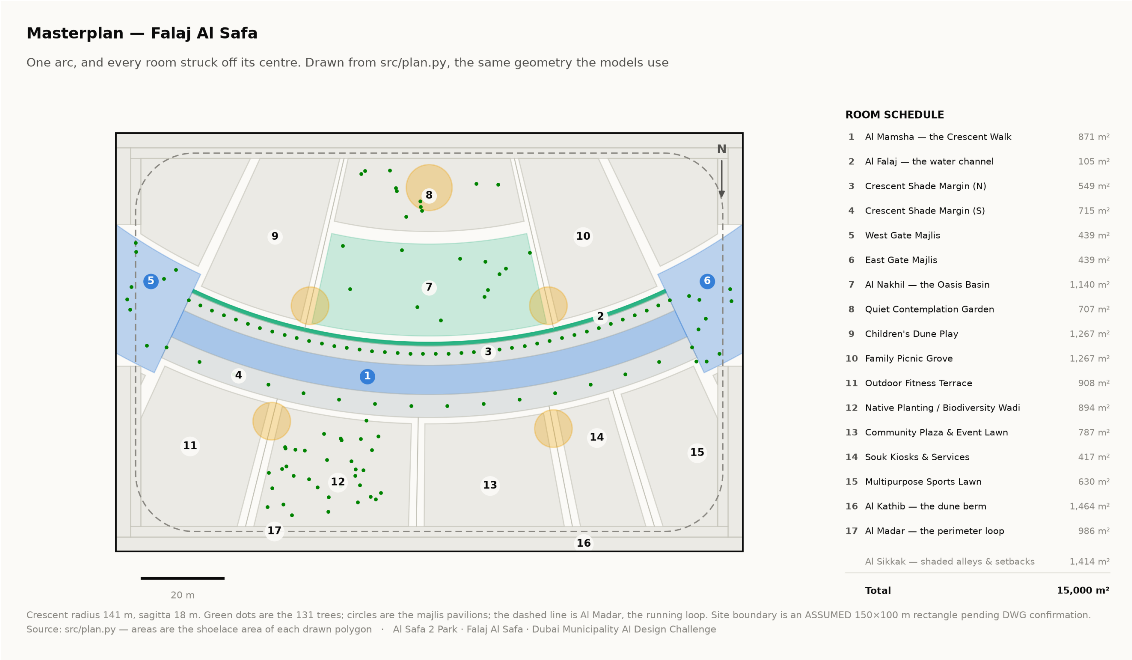
Upload slot 02 — Preliminary Design Masterplan
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The organising geometry

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

The arc has a chord of 138 m and a sagitta of 18 m, giving a radius of 141.25 m and a true arc length of 144.2 m. Those four numbers generate everything else in this report.



Masterplan with the numbered room schedule. Technical drawing — to scale.

2. Room schedule

Room	Category	Area (m²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%

Room	Category	Area (m²)	% of site
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

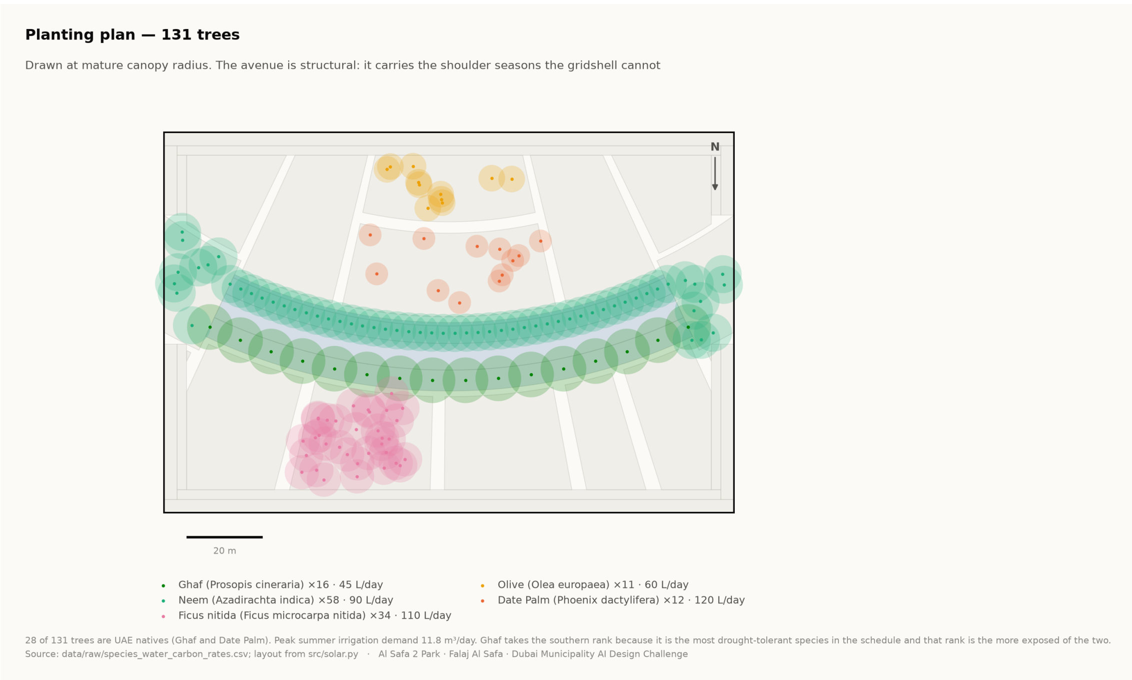
Total 15,000 m². Areas are shoelace areas of the drawn polygons, and the pipeline asserts that the schedule sums to the site area.

3. Where things are, and why

- **The hollow holds the lingering.** The Oasis Basin, the quiet garden, the children's play and the picnic grove all sit on the concave side, which the comfort model shows is measurably cooler than the convex face.
- **The convex face takes the active and the commercial.** The sports lawn, the community plaza and the souk sit south of the arc, where exposure matters less and evening use dominates.
- **The alleys are radii.** Every sikka is a radius of the same circle, so every room presents its face to the crescent rather than a corner.
- **The perimeter is a berm, not a fence.** Al Kathib is planted earth against the roads, doing three jobs at once — noise, glare and heat.

4. Site boundary — a stated assumption

The site is modelled as a 150 × 100 m rectangle. This is an **assumption** pending confirmation against the CAD file issued with the competition documents. Every area figure in this submission depends on it, and it is flagged here rather than left for a juror to discover.



Planting plan — 131 trees at mature canopy radius. Technical drawing — to scale.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.